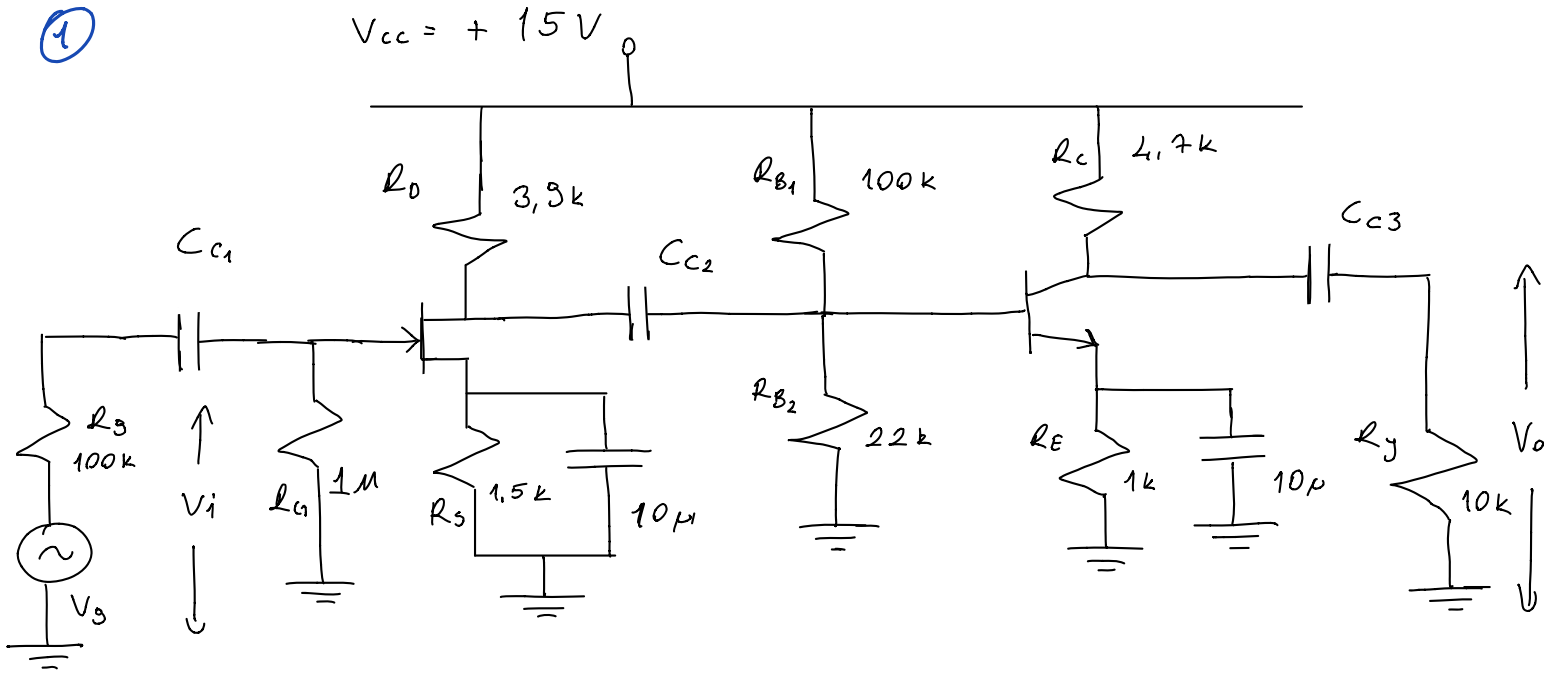


1)



$$I_{DSS} = 6mA, \quad V_p = -3V, \quad \beta = h_{fe} = h_{FE} = 200, \quad V_{BE} = 0.7V$$

$$1/h_{oe} = 50k, \quad C_{gs} = 3pF, \quad C_{ds} = 1pF, \quad C_{gd} = 3pF$$

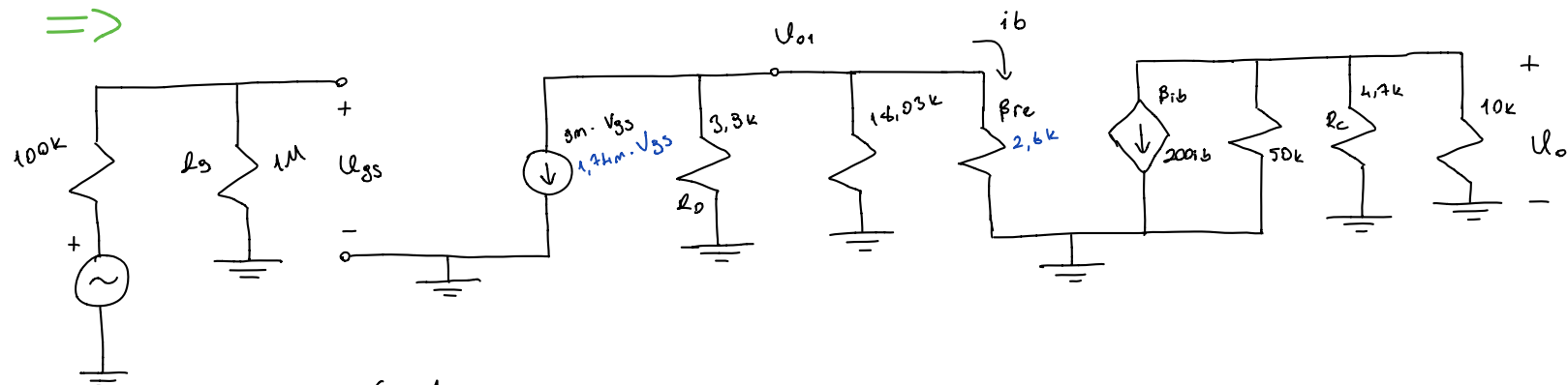
$$C_{be} = 80pF, \quad C_{bc} = 3pF, \quad C_{ce} = 6pF$$

a) $K_v = \frac{V_o}{V_i} = ?$

b) Alt kesim frekansının 10Hz olabilmesi için, $C_{c1}, C_{c2}, C_{c3} = ?$

c) Üst kesim frekansı nedir?

=>



$$g_m = 2 \frac{6mA}{3} (1 - 0.565) = 1.74m$$

a)

$$r_c = \frac{26mV}{I_{CQ}}$$

$$V_B = 15 \cdot \frac{22k}{122k} = 2.7V$$

$$V_E = 2.7 - 0.7 = \underline{2V}$$

$$I_{CQ} \cong I_{EQ}$$

$$I_{EQ} = \frac{2}{1k} = 2mA$$

$$r_c = 13\Omega$$

$$r_{ie} = 2.6k\Omega$$

$$A_{v1} = -g_m(3,3k // R_{i2}) \quad R_{i2} = 18,03k // 2,6k = 2,272k$$

$$A_{v1} = \underline{\underline{-2,498}}$$

$$A_{v2} = \frac{-200}{2,6k} \underbrace{(50k // 4,7k // 10k)}_{3,005k} = \underline{\underline{-231,15}}$$

$$A_v = A_{v1} \cdot A_{v2} = \underline{\underline{577,41}}$$

b)

$$C_{c1} \Rightarrow 10 = \frac{1}{2 \cdot \pi \cdot C_{c1} \cdot (100k + 1M)} \Rightarrow \underline{\underline{C_{c1} = 14,46 nF}}$$

$$C_{c2} \Rightarrow 10 = \frac{1}{2 \pi \cdot C_{c2} \cdot (2,27k + 3,9k)} \Rightarrow \underline{\underline{C_{c2} = 2,57 \mu F}}$$

$$C_{c3} \Rightarrow 10 = \frac{1}{2 \cdot \pi \cdot C_{c3} \cdot (4,7k + 10k)} \Rightarrow \underline{\underline{C_{c3} = 1,48 \mu F}}$$

c)

- FET -

$$C_{in} = C_{gs} + C_{gd}(1 - A_{v1}) + C_{w1}$$

$$C_{in} = 3pF + 3pF(1 + 2,498) + 0$$

$$C_{in} = \underline{\underline{13,49 pF}} \quad R_{in} = 1M // 100k \approx \underline{\underline{90,9 k\Omega}}$$

$$f_{cin} = \frac{1}{2 \cdot \pi \cdot 13,43 \times 10^{-12} \cdot 90,9 \cdot 10^3} = \underline{\underline{129,730 kHz}}$$

$$C_{out} = C_{ds} + C_{gs} \left(1 - \frac{1}{A_{v1}}\right) + C_{w2}$$

$$C_{out} = 1pF + 3pF \left(1 + \frac{1}{2,498}\right) + 0 \approx \underline{\underline{5,2 pF}}$$

$$R_{out} = 3,9k // 2,27 = \underline{\underline{1,434 k\Omega}}$$

$$f_{cout} = \frac{1}{2 \cdot \pi \cdot 5,2 \times 10^{-12} \cdot 1,434} = \underline{\underline{21,343 MHz}}$$

- BJT -

$$C_{in} = C_{be} + C_{bc}(1 - A_{v2}) + C_{w1}$$

$$C_{in} = 80pF + 3pF(1 + 231,15) + 0 \approx \underline{\underline{776,4 pF}}$$

$$R_{out} = 3,9k // 2,27 = \underline{\underline{1,434 k\Omega}}$$

$$f_{cin} = \frac{1}{2 \cdot \pi \cdot 776,4 \times 10^{-12} \cdot 1,434} = 143 kHz$$

$$C_{out} = C_{ce} + C_{bc} \left(1 - \frac{1}{A_{v2}}\right) + C_{w1}$$

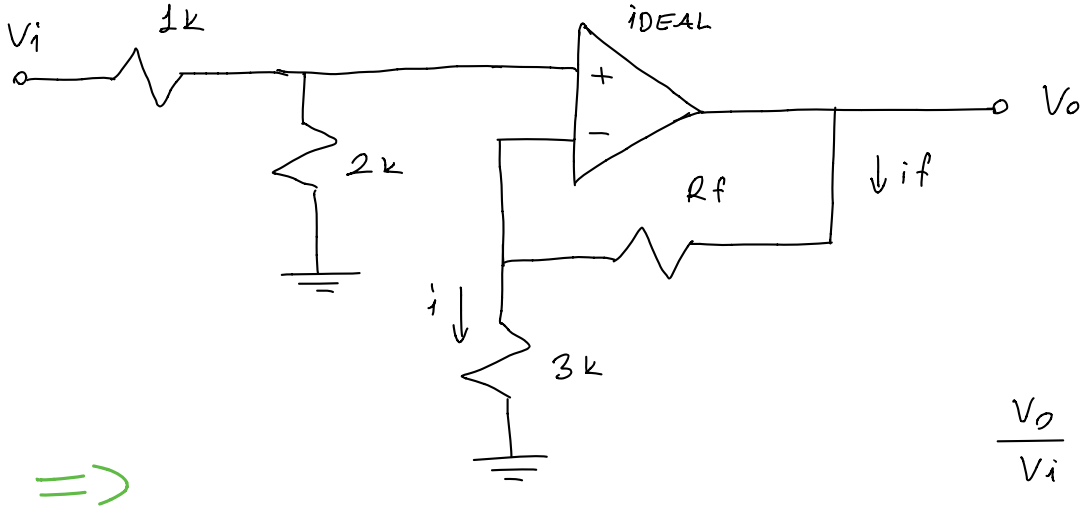
$$C_{out} = 6pF + 3pF \left(1 + \frac{1}{231,15}\right) + 0 \approx \underline{\underline{9,02 pF}}$$

$$R_{out} = 50k // 4,7k // 10k = \underline{\underline{3,005 k\Omega}}$$

$$f_{cout} = \frac{1}{2 \cdot \pi \cdot 9,02 \cdot 10^{-12} \cdot 3,005} = \underline{\underline{5,67 MHz}}$$

$$\underline{\underline{f_{cut} = 129,730 kHz}}$$

② Aşağıda verilen ideal OPAMP'ın kazancının 5 olabilmesi için R_f ne olmalıdır?



Gerilim Bölücü

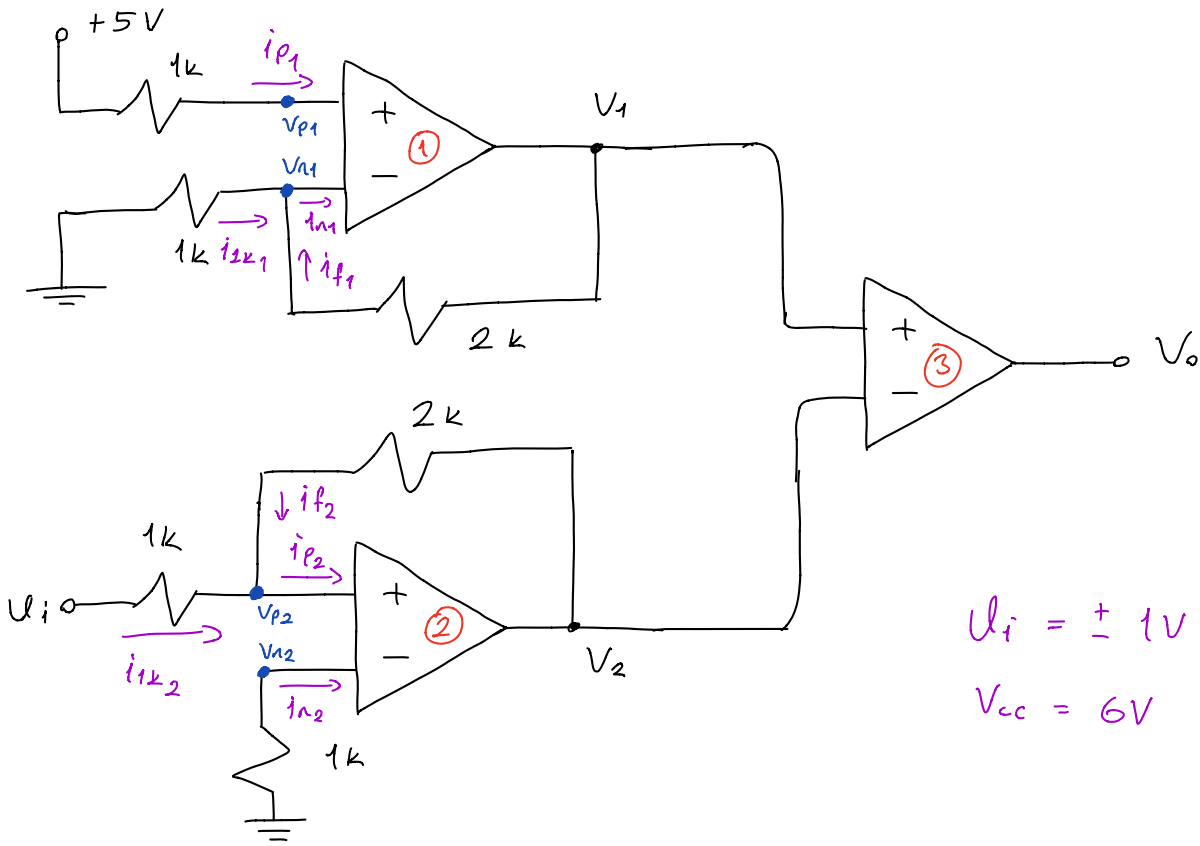
$$V_s = V_i \frac{2k}{1k + 2k} = \frac{2V_i}{3}$$

$$\frac{V_o}{V_s} = \frac{V_o}{\frac{V_i \cdot 2k}{3k}} \Rightarrow \frac{3}{2} \cdot \frac{V_o}{V_i} = \frac{3.5}{2} = \frac{15}{2} //$$

$$\frac{V_o}{V_s} = 1 + \frac{R_f}{R_i} \Rightarrow \frac{15}{2} = 1 + \frac{R_f}{3k} = \frac{33k}{2}$$

$$\Rightarrow \underline{\underline{R_f = 19,5k\Omega}}$$

③



$$U_i = \pm 1V \text{ (sinüzoidal)}$$

$$V_{cc} = 6V$$

Şekildeki devrede U_i , $\pm 1V$ sinüzoidal bir sinyal ve tüm Opamlar için $\pm V_{cc} = \pm 6V$ olduğuna göre, V_1 , V_2 ve V_o singallerini U_i 'ye göre ölçekli olarak çiziniz.

=>

1. OP-AMP

$$\frac{V_{n1}}{1k} + \frac{V_{n1} - V_1}{2k} = 0$$

$$V_{p1} = V_{n1} = 5V$$

$$10 + 5 - V_1 = 0 \Rightarrow V_1 = 15V \quad (V_{cc} = \pm 6V \text{ ise})$$

$$i_{p1} = i_{n1} = 0A$$

$$\Rightarrow \underline{\underline{V_1 = 6V}}$$

$$i_{k1} + i_{f1} = 0A$$

2. OP-AMP

$$\frac{V_i - V_{p2}}{1k} + \frac{V_2 - V_{p2}}{2k} = 0 \Rightarrow \frac{V_i}{1k} = \frac{-V_2}{2k}$$

$$i_{p2} = i_{n2} = 0A$$

$$V_{p2} = V_{n2} = 0V$$

$$\underline{\underline{V_2 = -2V_i \quad V}}$$

$$(V_i = \pm 1V \text{ (sinüzoidal)})$$

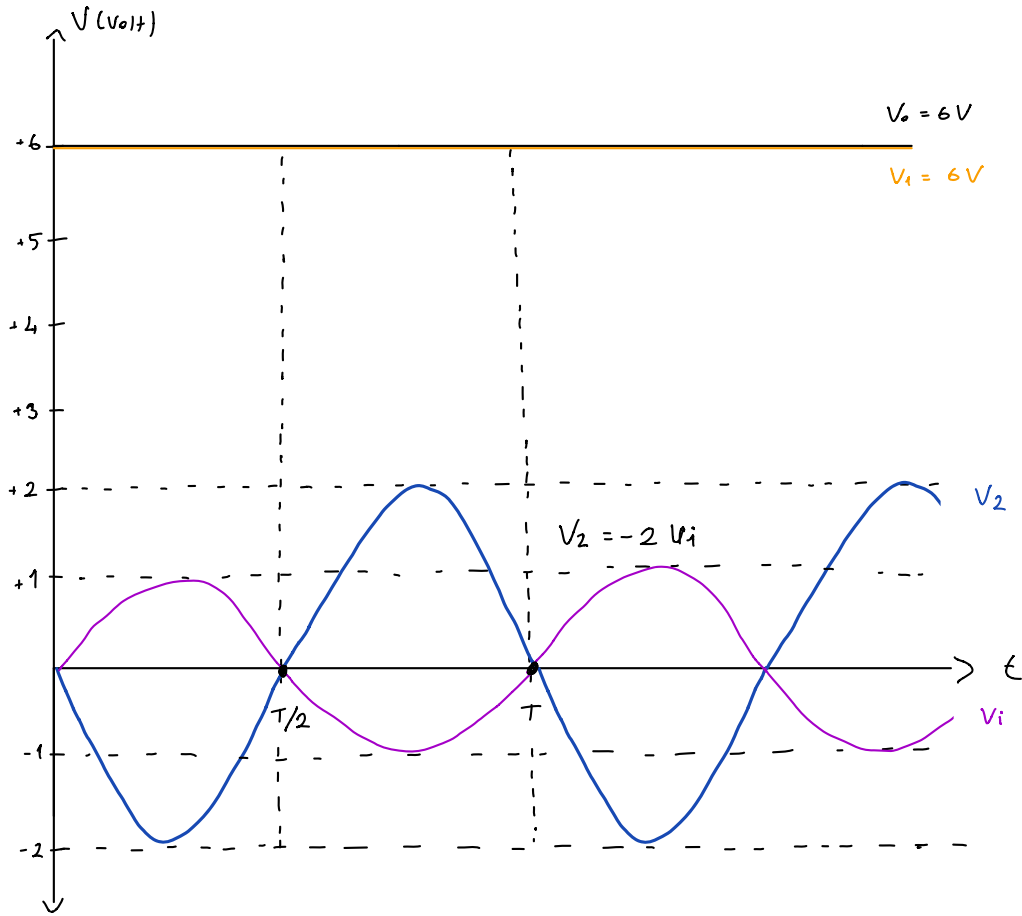
$$i_{k2} + i_{f2} = 0A$$

$$\Rightarrow \underline{\underline{V_2 = -2V \text{ (sinüzoidal)}}}$$

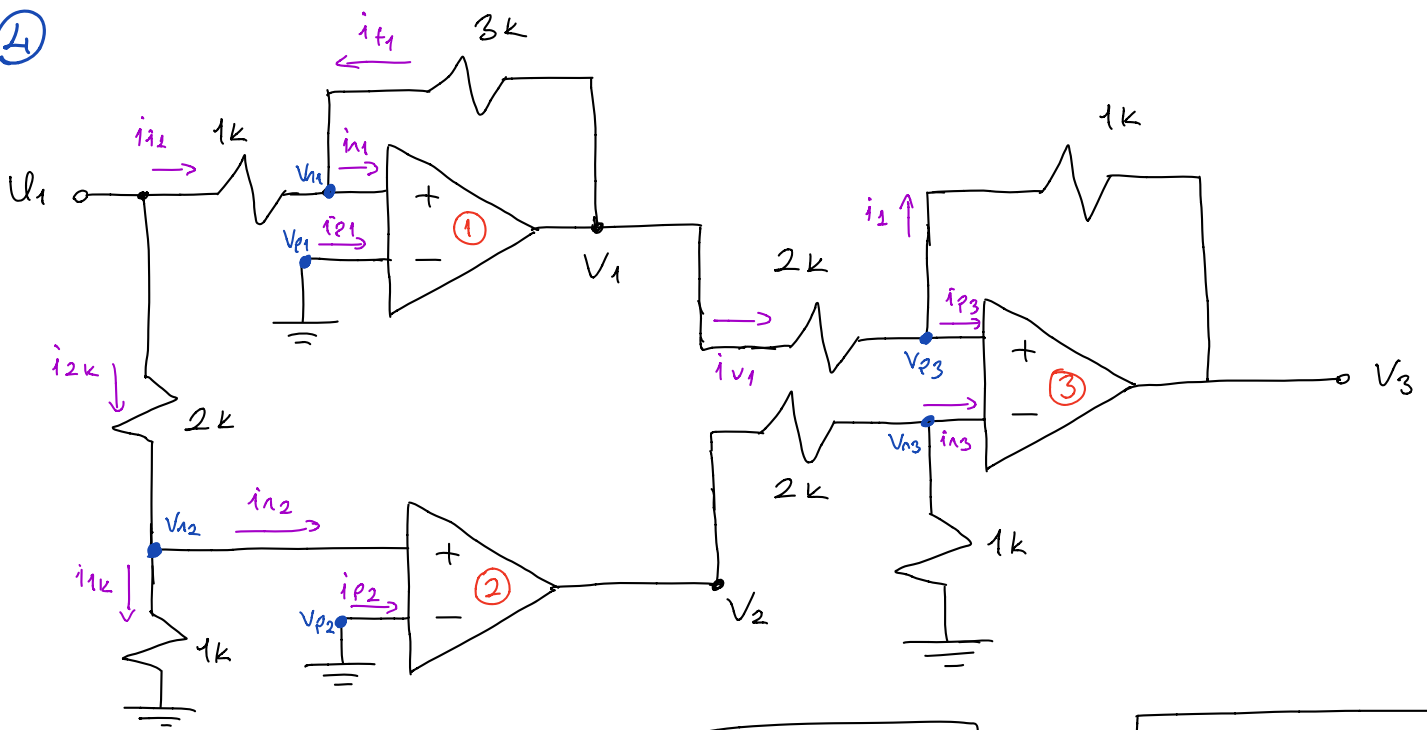
3. OP-AMP

$V_1 > V_2$ olduğundan dolayı

$$\underline{V_o = 6V}$$

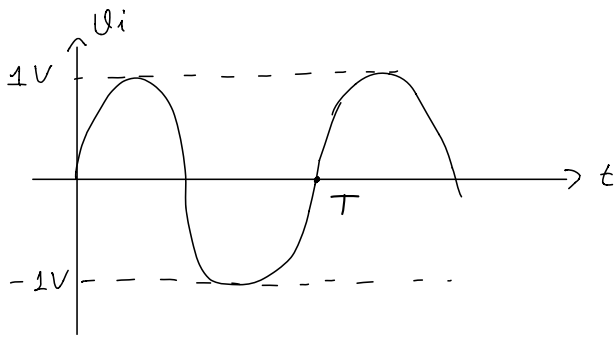


4)



$$+V_{cc} = \pm 12V$$

$$V_i = \pm 1V \sin t$$



Buna göre $V_1, V_2, V_3 = ?$
(Çiziniz)

$\Rightarrow$

1. OP-Amp

$$\left. \begin{aligned} V_{p1} &= V_{n1} = 0V \\ i_{n1} &= i_{p1} = 0A \\ i_{i1} + i_{f1} &= 0 \end{aligned} \right\}$$

$$\frac{0V}{V_{n1} - V_i} + \frac{0V}{V_{n1} - V_1} = 0 \Rightarrow \underline{V_1 = -3V_i}$$

2. OP-Amp

$$V_{n2} = \frac{V_i \cdot 1k}{2k + 1k} = \frac{V_i}{3}$$

$$V_{p2} = 0V$$

$$\left. \begin{aligned} V_{n2} &> V_{p2} \Rightarrow V_2 = \begin{cases} 12V, & V_{n2} < V_{p2} \\ 0V, & V_{n2} = V_{p2} \\ -12V, & V_{n2} > V_{p2} \end{cases} \end{aligned} \right\}$$

3. OP-AMP

$$V_{n3} = \frac{V_2 \cdot 1k}{2k + 1k} = \frac{V_2}{3}$$

$$i_1 = i_{v1}$$

$$V_{r3} = V_{p3}$$

$$\frac{\overbrace{V_{p3}}^{V_2/3} - V_3}{1k} = \frac{V_1 - \overbrace{V_{p3}}^{V_2/3}}{2k}$$

$$\Rightarrow \frac{V_2}{3} - V_3 = \frac{V_1}{2} - \frac{V_2}{6}$$

$$\Rightarrow V_3 = \frac{V_2 - V_1}{2}$$

